Estimating Dimensional Structure in Generative Psychometrics: Comparing PCA and Network Methods Using Large Language Model Item Embeddings

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Abstract

As large language models (LLMs) increasingly inform psychometric practice through item generation and semantic analysis, embedding-based approaches offer a pre-empirical pathway for assessing dimensional structure before human response data become available. However, the methodological choices used to recover dimensions from embedding spaces remain underexamined. The present study compared the performance of principal component analysis (PCA), which currently dominates generative psychometric applications, with network-based exploratory graph analysis (EGA) for estimating dimensional structure from item embeddings. Using Monte Carlo simulations across six personality facets nested within two domains, we generated 180 items per condition via LLM prompting and represented them as semantic embeddings. Dimensional recovery was evaluated across four analytic pipelines: Kaiser's rule with PCA extraction and varimax rotation, parallel analysis with PCA extraction and varimax rotation, EGA, and EGA combined with network-integrated item filtering. Results showed that PCA-based methods severely overestimated dimensionality and yielded poorer recovery of the generating structure. EGA demonstrated markedly improved performance relative to PCA-based approaches, whereas EGA combined with network filtering procedures achieved near-perfect dimensional accuracy and structural recovery. These findings indicate that network psychometric workflows provide more accurate structural representations than component-based approaches when analyzing embedding spaces, with direct implications for generative psychometric workflows in which item pools are evaluated prior to data collection. More broadly, the findings demonstrate that analytic choices made during pre-empirical assessment have substantive consequences for item selection, construct validity, taxonomic clarity, and downstream theoretical interpretation.

Keywords: scale development, large language models, item embeddings, principal component analysis, exploratory graph analysis, network psychometrics

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Introduction

Generative artificial intelligence (AI) is increasingly reshaping core activities in psychometrics, reflecting a broader shift toward treating language as a scalable behavioral trace for psychological science and cognitive modelling (Brickman & Oltmanns, 2025; Debelak & Stachl, 2025; Feuerriegel & Van Bavel, 2025; Jackson & Lindquist, 2022; Kjell, Kjell, & Schwartz, 2024; Van Quaquebeke & Banks, 2025). Large language models (LLMs) are now integrated into item-generation workflows, where they are used to generate, expand, and refine item pools, as well as to support translation and cross-cultural adaptation (Fyffe & Kaplan, 2024; Grobelny & Strozyk, 2025; Hernandez & Nie, 2023; Keane & McNaughton, 2025; Kowal & Kantrowitz, 2025; Lee, Son, & Jia, 2025; Perrin et al., 2025; Ruiz-Mendoza & Pedroza-Zúñiga, 2025; Russell-Lasalandra, Christensen, & Golino, 2024; Salah et al., 2025). These applications typically operate under structured human oversight or multi-agent quality-control pipelines (e.g., LLMs evaluating and critiquing each other's outputs) to reduce redundancy while anchoring item content in theory and lived experience. In parallel, transformer-based models trained on naturally occurring language are increasingly used to predict personality traits and related constructs from open-ended text. In this role, such models function as automated raters and, when fine-tuned, as scoring engines for assessment and psychological text-classification tasks (Bunt & Gillespie, 2025; Hommel, 2023; Kjell et al., 2024; Lee et al., 2025; Maharjan, Jin, Zhu, & Kenne, 2025). As these embedding-based approaches move beyond proof-of-concept demonstrations toward routine application, however, fundamental questions remain unresolved, particularly regarding how to reliably recover dimensional structure from language-based representations.

Text embeddings sit at the center of many of these developments, providing a general mechanism for transforming language into quantitative data suitable for psychological analysis (Brickman & Oltmanns, 2025; Feuerriegel & Van Bavel, 2025; Jackson & Lindquist, 2022). An embedding model maps a unit of text, such as an item stem, construct label, or self-description, into a dense vector in a high-dimensional space. Through this mapping, scale items and survey questions can be treated as numeric representations of their semantic content (Fang & Oberski, 2022; Russell-Lasalandra et al., 2024; Stanton & Sang, 2024). These vectors are learned from large text corpora such that semantically similar texts tend to occupy nearby locations, whereas dissimilar texts lie farther apart, yielding a geometric space that encodes graded semantic relations among words, items, and other psychologically relevant concepts (Fang & Oberski, 2022; Golino, 2025; Hussain, Binz, Mata, & Wulff, 2024). Once texts have been embedded, researchers can compute similarity metrics between items, derive distance-based neighborhoods and clusters, or use embeddings as inputs to supervised models that classify items into constructs (Fang & Oberski, 2022; Fyffe & Kaplan, 2024). In this sense, language is rendered into numerical form that can be analyzed using many of the same tools applied to Likert-type response data, including similarity matrices, matrix factorizations, clustering algorithms, and predictive models of scale characteristics (Hommel & Arslan, 2025; Hussain et al., 2024; Stanton & Sang, 2024).

Capitalizing on evidence that embedding-based distances and configurations can mirror patterns observed in human response data (Fang & Oberski, 2022; Hommel & Arslan, 2025; Hussain et al., 2024; Russell-Lasalandra et al., 2024; Stanton & Sang, 2024), researchers have applied embedding-based approaches to scale abbreviation without collecting response data, to mapping nomological neighborhoods across hundreds of thousands of indicators, and to pre-data-collection diagnostics of content and construct validity (Fyffe & Kaplan, 2024; Kilmen & Bulut, 2025a; Larsen & Edmondson, 2025; Ravenda & Raballo, 2025; Wulff & Mata, 2025a, 2025b).

Complementing these diagnostics, LLMs can also be prompted to emulate specific profiles and generate synthetic respondents whose answers approximate human response patterns, producing pseudo-response data that can be analyzed using conventional psychometric models (Cipriani & Grassini, 2025; Niszczota & Misiak, 2025; Rosenfelder & Gilead, 2025). Together, these tools have the potential to make scale development, validation, and structural analysis more efficient and less dependent on large human samples during early stages of instrument development. Realizing this potential, however, depends critically on the methods used to recover dimensional structure from embedding spaces, a choice that has received limited scrutiny despite its downstream implications for construct validity and measurement quality.

Current approaches to this structural analysis problem rely heavily on methods that may be poorly suited to the task. In embedding-based psychometric applications, dimensional structure assessment is dominated by principal component analysis (PCA) applied to cosine- or correlation-based similarity matrices derived from item, adjective, or construct embeddings (Bhandari & Pardos, 2024; Cutler & Condon, 2023; Huang & Tong, 2025; Jung & Seo, 2025; Linden, Cutler, Van der Linden, & Dunkel, 2025; Milano & Marocco, 2025; Teitelbaum & Simchon, 2025). The dominance of PCA in these workflows reflects a long historical tradition in fields such as personality and psychopathology, where PCA has served as the default tool for deriving multilevel structure from large item pools (Ashton & Goldberg, 2004; Forbes & Clark, 2025; Goldberg, 2013; Saucier & Iurino, 2020). Proponents of PCA argue that it offers advantages due to its mathematical simplicity, reduced susceptibility to problematic solutions, and absence of score indeterminacy (Sellbom & Goretzko, 2023a). However, a long-standing methodological literature raises concerns about relying on PCA as a primary basis for determining the dimensional structure of psychological variables, particularly when paired with eigenvalue-greater-than-one or scree-plot heuristics (Goretzko & Bühner, 2021; Kaçak & Kılıç, 2025; Sellbom & Goretzko, 2023b; Widaman, 1993; Widaman, 2018).

Conceptually, PCA is a descriptive decomposition of total variance rather than a representation of common variance. As a result, components maximize variance in observed variables without separating common, unique, and error variance, providing no guarantee that the resulting dimensions correspond to psychological attributes (Goretzko & Bühner, 2021; Widaman, 2018). Simulation studies indicate that, under conditions typical of psychological data, including low to moderate communalities, few indicators per dimension, and ordinal-level measurement, PCA tends to overestimate factor loadings and can yield biased recovery of the generating structure (Kaçak & Kılıç, 2025; Widaman, 1993). Moreover, the PCA framework is poorly suited to address issues that are particularly relevant for generative AI-based psychometric pipelines, such as item redundancy, structural instability (Russell-Lasalandra et al., 2024), and hierarchical assessment (Samo & Christensen, 2025). Taken together, these conceptual and empirical critiques suggest that PCA-based dimensional solutions, while convenient and historically entrenched, provide an uncertain foundation for inferences about dimensional structure. This concern becomes especially salient when PCA is used to characterize the geometry of embedding spaces and guide downstream modeling choices in generative AI-based psychometrics.

Generative AI Psychometrics with Network Models

Network modeling conceptualizes psychological variables as nodes in a network, with conditional associations between variables represented as edges, typically estimated using Gaussian graphical models estimated via regularized partial correlations, or estimated via triangulated maximally filtered graphs (Christensen, Garrido, Guerra-Peña, & Golino, 2024; Golino, Shi, et al., 2020). Within this framework, Exploratory Graph Analysis (EGA) integrates sparse network estimation with community detection algorithms, such as Walktrap or Louvain, to identify clusters of densely connected nodes corresponding to latent or emergent dimensions (Christensen, Golino, Abad, & Garrido, 2024; Cosemans, Rosseel, & Gelper, 2022; Golino, Thiyagarajan, et al., 2020; Jimenez & Garrido, 2025; Samo & Christensen, 2025). Simulation studies demonstrate that EGA recovers the number of dimensions with high accuracy across a wide range of factor structures and data types, often matching or outperforming traditional factor-retention methods for both continuous and categorical indicators (Cosemans et al., 2022; Golino, Thiyagarajan, et al., 2020).

Building on EGA's core framework, several specialized network-based tools have been developed to address challenges that are amplified in large item pools. Bootstrap EGA (bootEGA) extends the base approach by repeatedly re-estimating networks across bootstrap samples, yielding indices of structural consistency and item stability (Christensen & Golino, 2021). To address item redundancy, Unique Variable Analysis (UVA) integrates network estimation with weighted topological overlap to identify locally dependent items prior to dimensionality assessment, demonstrating strong performance across data types and favorable comparisons to alternative diagnostics (Christensen & Golino, 2023). For interpretation, network loadings provide factor-loading-like coefficients that yield dimension scores whose correlation patterns reflects the generating correlations among dimensions, without requiring rotational decisions (Christensen & Garrido, 2025). When response styles or wording effects are present, random intercept EGA partials out these sources of variance prior to network estimation, improving dimensionality recovery (Garcia-Pardina, Abad, Christensen, Golino, & Garrido, 2024). Finally, hierarchical EGA enables accurate recovery of both general and group factors from item-level data (Jiménez et al., 2023).

These developments converge in Taxonomic Graph Analysis (TGA), a comprehensive framework for recovering hierarchical construct structures from large item pools (Samo & Christensen, 2025). TGA integrates the tools described above into a bottom-up workflow: UVA identifies redundant items, random intercept EGA mitigates wording effects, EGA with consensus clustering estimates dimensionality across successive hierarchical levels, bootEGA evaluates structural stability, and network loadings quantify indicator strength and derive scores for higher-order structure estimation. This integrated approach yields multilevel taxonomies that coherently link item-level facets, traits, and meta-traits within a single hierarchy (Christensen & Golino, 2021; Christensen & Garrido, 2025; Christensen & Golino, 2023; Samo & Christensen, 2025).

The Automatic Item Generation with Network-Integrated Evaluation (AI-GENIE) framework (Russell-Lasalandra et al., 2024) extends this taxonomic logic to generative AI psychometrics by coupling LLM-based item generation with embedding models and applying the same UVA-EGA-bootEGA pipeline to embedding-based similarity structures. This approach identifies nonredundant, structurally stable items from large LLM-generated pools and produces concise scales whose structural properties align with established trait measures, illustrating how network psychometrics can automate early-stage scale development in generative AI contexts (Russell-Lasalandra et al., 2024). To date, however, no study has systematically compared these network-based approaches with the predominant PCA pipeline for recovering dimensional structure from embedding-derived similarity matrices.

The Present Study

The present study addresses this methodological gap by directly comparing PCA and network approaches to dimensional structure estimation in embedding-based psychometrics. We conducted a Monte Carlo simulation in which LLMs generated item pools for multiple psychological constructs. Text embeddings were then derived from these AI-generated items to construct cosine similarity matrices analogous to those used in current embedding-based structural analysis pipelines (Russell-Lasalandra et al., 2024). Dimensional structure was estimated from these embedding-based similarity matrices using PCA with traditional retention criteria, including the eigenvalue-greater-than-one criterion (**K1**) or Kaiser's rule (Kaiser, 1960) and parallel analysis [PA; Horn (1965)], as well as the EGA framework.

The accuracy of each method was evaluated based on its ability to recover the generating dimensional structure. Because AI-GENIE's redundancy and stability filtering procedures are inherently network-based, post–item-filtering analyses were conducted exclusively within the EGA framework. Specifically, we examined how filtering via `UVA` and `bootEGA` affected dimensional structure recovery within the network approach. Given the conceptual and empirical limitations of PCA documented in the psychometric literature

(Goretzko & Bühner, 2021; Kaçak & Kılıç, 2025; Sellbom & Goretzko, 2023b; Widaman, 1993; Widaman, 2018), we hypothesized that EGA would demonstrate superior accuracy in recovering dimensional structure from embedding spaces and would yield more informative item-level diagnostics following network-based filtering. These comparisons have direct implications for the methodological foundations of generative AI-based scale development, where early structural inferences guide downstream validation and application contexts.

Method

Simulation Design

The accuracy of the dimensional structure estimates was evaluated using Monte Carlo simulation. The GPT-4o model (OpenAI, 2023a) under the default nucleus sampling ($\mathbf{p} = 1$) and temperature (1.0) was used to generate item pools measuring six personality facets taken from two domains (Irwing & Booth, 2024): (1) the facets of achievement striving, perseverance, and self-efficacy for the domain of **conscientiousness**, and (2) the facets of abstract thinking, aesthetics, and introspection for the domain of **openness to experience**. A target pool of 90 items per domain was set (180 total), which the LLM distributed approximately equally across the specified facets. We chose to generate a large number of items per facet (30) to mimic expected procedures in real-life scenarios of AI-aided test development where 3, 4 or more times the intended final number of items are generated initially (Koweszko & Silczuk, 2025; Lee & Yao, 2023). A total of 500 replications were simulated.

Item Generation and Prompt Engineering

Effective use of LLMs for psychological assessment requires systematic prompt engineering—the deliberate design and optimization of input prompts to guide model responses toward accurate, relevant, and coherent output (Chen, Zhang, Zheng, & Luo, 2025). The quality of prompts plays a significant role in determining the consistency and viability of generated content (Jin, Cheng, Shen, Chen, & Ren, 2022; White, Hays, Fu, Spencer-Smith, & Schmidt, 2023), making careful prompt construction essential for psychometric applications. This study employed a comprehensive set of prompt engineering techniques following the AI-GENIE approach (Russell-Lasalandra et al., 2024): role prompting, clear and precise prompting, and few-shot prompting. Such a multi-faceted approach is necessary because the vast universe of potential LLM responses can lead to highly variable outputs; detailed, unambiguous prompts help constrain the model toward generating content aligned with specific research goals (Ekin, Yilmaz, Erol, & Karagöz, 2023; Giray, 2023; see: OpenAI, 2023b).

The pipeline began with role or persona prompting, which assigns a specific expert identity to guide the model's responses. Research demonstrates that priming a model with a knowledgeable persona improves performance by producing more coherent, domain-relevant responses and reducing generalization errors (Zheng et al., 2023). When focused on a particular professional role, the model generates outputs that are more logically consistent and tailored to the specific context. Accordingly, the model was informed: *"You are an expert psychometrician and test developer specializing in personality assessment."*

Second, clear and precise prompting was used to specify the nature of the task, including the target domains, facets, and operational definitions for the items to be generated (Irwing & Booth, 2024). Providing relevant context and detailed constraints helps the model understand the prompt's purpose and produce appropriately targeted content. The prompt stated: *"Your task is to create high-quality, psychometrically robust items for a personality inventory measuring the following Big Five traits: Conscientiousness and Openness to Experience. Conscientiousness reflects self-efficacy (Propensity to hold the subjective perception that one is capable of performing), achievement-striving (Propensity to be ambitious and goal-oriented), and perseverance (Propensity to continue with and finish a task despite obstacles). Openness reflects introspection (Propensity to examine one's thoughts,* *feelings, motives, and behaviour), aesthetics (Enjoyment of artistic and aesthetic activities), abstract-thinking (Propensity to explore and discuss abstract ideas)."*

Third, detailed formatting guidelines were provided to ensure predictable, extractable output. LLMs attend closely to emphasized instructions using atypical casing, and repeating key directives improves adherence (Amatriain, 2024; Google Developers, 2025). The formatting instructions were: *"Generate psychometrically reliable and valid personality assessment items for each trait's example item characteristics. Do NOT mix characteristics in single items. EACH item should represent ONE and ONLY ONE characteristic in ONE trait. EACH item should be ONE sentence. You should format your output using: [trait]: [item content]. Do NOT number or add ANY other text to your response. ONLY output the traits EXACTLY as provided. ONLY output the trait and item content – NOTHING else."*

Finally, few-shot prompting was employed by providing five to six example items for each facet. This technique encourages quality and consistency by allowing the model to discern patterns from exemplars, improving performance and enabling nuanced generalization (Brown et al., 2020). Including multiple trait-relevant examples also helps ensure that generated items capture important aspects of the target construct rather than being unidimensional. The example items were drawn from the Facet-level Multidimensional Assessment of Personality (Facet MAP) version I (Irwing & Booth, 2024), with all inventory items for the selected facets included in the prompt.

Data Structure Assessment

The dimensional structure of the generated items was assessed using cosine similarity matrices derived from item embeddings (Kilmen & Bulut, 2025b). Item embeddings were obtained using OpenAI's *text-embedding-3-small* model (OpenAI, 2024). Three analytic procedures were employed to evaluate the dimensional structure of the resulting item pools:

K1-PCA-Varimax: Commonly referred to as the "Little Jiffy" procedure (Kaiser, 1970), this approach combines the K1 dimensionality rule with PCA extraction and varimax orthogonal rotation. For over five decades, this procedure has been amongst the most widely used methods for dimensional structure assessment in the behavioral sciences (Fabrigar, Wegener, MacCallum, & Strahan, 1999; Howard & Henderson, 2023; Howard & O'Sullivan, 2025; Izquierdo, Olea, & Abad, 2014; Kaiser, 1970; Preacher & MacCallum, 2003).

PA-PCA-Varimax: this procedure represents a modification of the Little Jiffy approach in which the K1 rule is replaced with parallel analysis, one the best-performing and most extensively validated dimensionality estimation methods (Garrido & Ponsoda, 2013; Golino, Shi, et al., 2020; Goretzko, 2025; Xia, 2021). This modification is critical, as the K1 rule is known to perform poorly at the sample level, particularly in conditions with a large number of variables per dimension (Auerswald & Moshagen, 2019; Golino, Thiyagarajan, et al., 2020; Martínez Ramos & Rondón Sepúlveda, 2025). PA was computed using the component eigenvalues, the mean aggregation criterion, and 100 simulated data matrices per analysis.

EGA: the EGA approach was included as a modern, state-of-the-art alternative for dimensional structure assessment on cosine similarity matrices of item embeddings (Russell-Lasalandra et al., 2024). EGA was estimated using the graphical least absolute shrinkage and selection operator with extended Bayesian information criterion for model selection (EBICglasso) (Chen & Chen, 2008; Epskamp & Fried, 2018; Friedman, Hastie, & Tibshirani, 2008), with variable clustering determined via the Walktrap algorithm (Brusco & Watts, 2023; Pons & Latapy, 2006).

Item Filtering

AI-based psychometric pipelines typically begin with the generation of a large number of initial items per dimension that vary in psychometric quality and may include overlapping content. Consequently, identifying and removing redundant or structurally unstable items is essential for obtaining robust dimensional structure estimates. In the present study, item filtering was conducted using the *Network-Integrated Evaluation* approach included in AI-GENIE (Russell-Lasalandra et al., 2024). The filtering procedure began with UVA (Christensen & Golino, 2023), which was used to automatically detect and remove redundant items from the initial item pool. EGA was then applied to estimate the dimensional structure of the retained items. In the final step, bootEGA (Christensen & Golino, 2021) was used to evaluate item stability within each dimension, resulting in the removal of items that did not meaningfully contribute to dimensional stability. A cutoff value of 0.20 was used for UVA to identify item redundancy (Christensen & Golino, 2023; Samo & Christensen, 2025), and a minimum stability threshold of 0.75 was required for item retention in bootEGA (Russell-Lasalandra et al., 2024; Samo & Christensen, 2025). The bootEGA procedure employed the same analytical methods as EGA and was iteratively repeated until all retained items demonstrated adequate stability. Each bootEGA run was based on 500 parametric bootstrap samples generated from the matrix of cosine similarities (Christensen & Golino, 2021; Samo & Christensen, 2025). Both UVA and bootEGA were computed using a sparse embedding matrix rather than the full embedding matrix (Russell-Lasalandra et al., 2024).

Performance Criteria

Dimensionality estimation accuracy was evaluated using the percentage of correct estimates (PC) and the mean bias error (MBE). The formulas for these criteria are provided in Equations 1 and 2:

$$\mathrm{PC} = \frac{\sum C}{N_r} \times 100, \quad \text{where } C = \begin{cases} 1 & \text{if } \hat{\theta} = \theta \\ 0 & \text{if } \hat{\theta} \neq \theta \end{cases} \tag{1}$$

$$\mathrm{MBE} = \frac{\sum(\hat{\theta} - \theta)}{N_r} \tag{2}$$

where N_r is the number of replicates being aggregated, $\hat{\theta}$ is the number of dimensions estimated, and θ is the number of generating dimensions.

The PC metric is bounded between 0 and 100, where 0 denotes **no accuracy** and 100 indicates **perfect accuracy**. By contrast, a value of 0 on the MBE metric reflects the absence of bias, whereas negative and positive values indicate underestimation and overestimation of the number of generating dimensions, respectively.

Recovery of the dimensional structure was assessed using the *normalized mutual information* (NMI, see Equation 3) index (Danon & Arenas, 2005). NMI is an information-theoretic measure that compares the composition of estimated and generating structures in terms of item groupings. The index ranges from 0, indicating completely dissimilar structures, to 1, indicating identical structures, and can be computed for solutions that differ in the number of dimensions. NMI is computed from a confusion matrix, in which rows represent the generating dimensions and columns represent the estimated dimensions. Each entry denotes the number of items belonging to generating dimension that are assigned to estimated dimension. For PCA, items were assigned to the dimension on which they exhibited the highest loading, whereas EGA assigns items to dimensions as part of its clustering procedure. Based on the information-theoretic concept of mutual information, NMI is then defined as:

$$\mathrm{NMI} = \frac{-2\sum_{i=1}^{C_A}\sum_{j=1}^{C_B} N_{ij}\log\left(\frac{N_{ij}N}{N_{i.}N_{.j}}\right)}{\sum_{i=1}^{C_A} N_{i.}\log\left(\frac{N_{i.}}{N}\right) + \sum_{j=1}^{C_B} N_{.j}\log\left(\frac{N_{.j}}{N}\right)} \tag{3}$$

where C_A is the number of generating dimensions and C_B is the number of estimated dimensions.

Statistical Software

All statistical analyses were conducted in R (version 4.5.2). Eigenvalues required for computing the K1 rule were obtained using the base R function *eigen*. PA and PCA with varimax rotation were performed using the *psych* package (version 2.5.6) via the *fa.parallel* and *principal* functions, respectively. EGA, bootEGA and UVA were conducted using the *EGAnet* package (version 2.4.0) with the corresponding functions. NMI was computed using the *aricode* package (version 1.0.3) via the *NMI* function. All R code and simulation results are available at [OSF repository URL].

Results

The LLM was prompted to generate 90 items for conscientiousness and 90 for openness to experience, corresponding to 30 items per facet. In practice, however, LLMs generate and distribute items approximately rather than exactly. To illustrate the resulting distributions, Table 1 reports the mean number of items generated and retained for each facet across simulation replicates. As shown in the table, all the facets had mean item counts close to the intended value of 30, ranging from a low of 27.54 for abstract thinking to a high of 34.27 for introspection.

The performance of the analytic procedures in dimensional structure estimation is summarized in Table 2. The poorest performance was observed for K1-PCA-Varimax, which severely overestimated the number of dimensions (MBE = 15.06) and yielded the weakest recovery of the dimensional structure (NMI = 0.82). Although PA-PCA-Varimax improved both structural recovery (NMI = 0.88) and dimensionality estimation relative to K1-PCA-Varimax, it still markedly overestimated the number of dimensions (MBE = 7.37). Consistent with these findings, the PC was 0.0% for both PCA-based methods, indicating that neither correctly recovered the generating number of dimensions in any simulation replicate. In contrast, EGA demonstrated strong overall performance, with a PC of 43.4%, only a small upward bias in dimensionality estimation (MBE = 0.83), and substantially more accurate recovery of the dimensional structure (NMI = 0.95).

Following the Network-Integrated Evaluation filtering procedure, approximately

Table 1

Number of items generated and retained per personality facet

	Generated Items		Retained Items		
Dimension / Facet	Generated Items Mean	SD	Retained Items Mean	SD	% Retained
Achievement striving	30.40	1.19	17.31	3.67	56.9%
Perseverance	29.13	1.50	16.90	3.80	58.0%
Self-efficacy	32.63	1.55	22.28	4.23	68.3%
Abstract thinking	27.54	1.99	16.41	3.85	59.6%
Aesthetics	30.02	1.68	17.51	3.83	58.3%
Introspection	34.27	2.23	20.97	4.28	61.2%

Note. Retained items are those that remained after the Network-Integrated Evaluation filtering procedure.

one-half to two-thirds of the items were retained across the facets (Table 1). Retention rates ranged from a low of 56.9% obtained for achievement striving to a high of 68.3% for self-efficacy. Notably, the standard deviation of retained item counts was roughly twice that of the generated item counts, indicating increased variability across replicates as a consequence of the filtering procedure.

As shown in Table 2, applying EGA to the filtered item pools substantially improved dimensional structure recovery. With respect to dimensionality estimation, the PC increased to 95.6%, bias was eliminated (MBE = 0.00), and recovery of the dimensional structure was nearly perfect (NMI = 0.99). To further illustrate these results, Figure 1 displays the distributions of MBE and NMI across the 500 simulation replicates.

Table 2

Performance of analytic procedures in dimensional structure estimation

Analytic procedure	PC	MBE	NMI
K1-PCA-Varimax	0.0%	15.06	0.82
PA-PCA-Varimax	0.0%	7.37	0.88
EGA	43.4%	0.83	0.95
EGA (filtered)	95.6%	0.00	0.99

Note. K1 = Kaiser's rule; PCA = principal component analysis; Varimax = orthogonal varimax rotation; EGA = exploratory graph analysis. Item pools were filtered using the Network-Integrated Evaluation procedure.

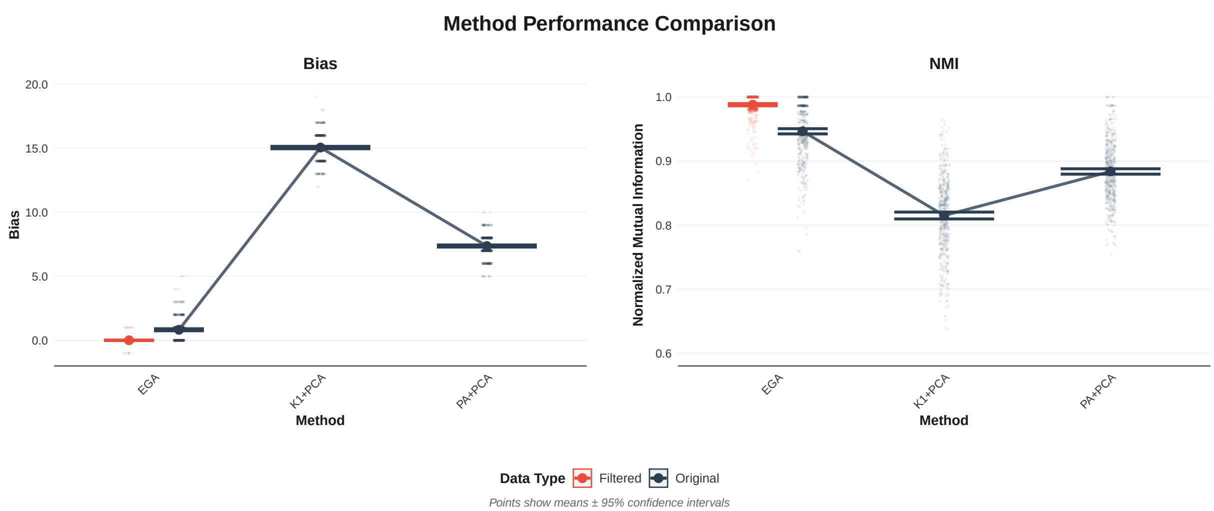

Figure 1. Performance comparison of analytic procedures for dimensional structure estimation. Panels show mean bias error (MBE), normalized mutual information (NMI), and estimated dimensionality for original and filtered item pools. Points represent means with 95% confidence intervals.

Discussion

The present study compared PCA and network-based approaches for estimating dimensional structure from LLM item embeddings, addressing a critical gap in the rapidly evolving field of generative AI psychometrics. Our primary aim was to evaluate whether the predominant reliance on PCA in embedding-based applications (Bhandari & Pardos, 2024; Cutler & Condon, 2023; Huang & Tong, 2025; Jung & Seo, 2025; Linden et al., 2025; Milano & Marocco, 2025; Teitelbaum & Simchon, 2025) is justified, or whether network methods such as Exploratory Graph Analysis and the related AI-GENIE pipeline (Russell-Lasalandra et al., 2024) offer meaningful advantages for this specific psychometric task. The results revealed substantial and consequential differences in favor of the network approach across multiple criteria, including dimensionality estimation accuracy and coherence of item clustering. These differences have important implications for the validity of dimensional structures recovered from embedding spaces and, by extension, for downstream applications that depend on accurate early-stage structural characterization of psychological constructs.

Main Findings

Of the two PCA-based pipelines, the popular "Little Jiffy" approach (Kaiser, 1970), which combines Kaiser's eigenvalue-greater-than-one rule (K1) with PCA extraction and varimax rotation (K1-PCA-Varimax), displayed the poorest performance, with severe overestimation of dimensionality and the weakest recovery of the generating structure. These results are consistent with a long line of research showing that the K1 rule performs poorly in finite samples, exhibiting a pronounced tendency toward overextraction (Auerswald & Moshagen, 2019; Golino, Thiyagarajan, et al., 2020; Martínez Ramos & Rondón Sepúlveda, 2025), and that PCA is a data reduction method that is not recommended for identifying dimensions arising from common variance among variables (Widaman, 1993). Proponents of PCA have argued that it offers procedural advantages, including mathematical simplicity, reduced susceptibility to improper solutions, and absence of score indeterminacy (Sellbom & Goretzko, 2023a). Although these arguments may appear compelling given the method's widespread adoption and default status in statistical software, the present findings, consistent with decades of methodological research, demonstrate that such conveniences come at the cost of substantial systematic distortions in dimensional structure estimation.

The findings also showed that substituting the K1 rule with parallel analysis (PA) in the PA-PCA-Varimax pipeline yielded meaningful improvements in both dimensionality estimation and structural recovery, consistent with PA's well-established superiority over Kaiser's rule in traditional psychometric contexts (Garrido & Ponsoda, 2013; Goretzko, 2025). Despite this improvement, however, PA-PCA-Varimax continued to markedly overestimate the number of dimensions and yielded less accurate structural recovery than EGA. This persistent overextraction, even with one of the most extensively validated and recommended retention methods in psychometrics, underscores fundamental limitations in applying traditional dimensionality workflows to LLM-generated item pools and embedding-based representations.

Eigenvalue-based dimensionality methods such as PA are more susceptible than network approaches to nuisance variance arising from minor dimensions (Jimenez & Garrido, 2025). In generative psychometrics, such nuisance variance can stem from multiple sources, including item redundancy and the emergence of narrow subdimensions driven by the large number of items generated by LLMs. In addition, large LLM-generated item pools are more likely to contain solitary items, defined as items that do not load saliently on any dimension. These items can inflate the eigenvalues of noise dimensions, thereby increasing the likelihood that they are retained by methods such as PA (Turner, 1998). Under these conditions, PA may yield dimensionality estimates that conflate major and minor dimensions, resulting in systemic overextraction (Garcia-Pardina et al., 2024).

In contrast, EGA demonstrated substantially superior performance in recovering dimensional structure from unfiltered embedding matrices. The network approach correctly identified the generating number of dimensions in nearly half of all replicates, whereas the PCA-based approaches never provided correct dimensionality estimates, and did so with only a slight upward bias and high structural recovery accuracy. These results extend prior simulation evidence supporting EGA's robust performance in traditional response-based data (Cosemans et al., 2022; Golino, Thiyagarajan, et al., 2020; Jimenez & Garrido, 2025) to the novel context of embedding-derived similarity matrices, demonstrating that the method's advantages persist when applied to semantic representations produced by contemporary LLMs. Unlike eigenvalue-based methods, EGA derives dimensional structure from sparse, regularized partial correlation networks and community detection algorithms that identify clusters of variables remaining conditionally dependent after accounting for all other variables in the system. This property makes EGA less susceptible to nuisance variance arising from minor dimensions, which are less likely to disrupt the substantive clustering of the variables in network representations (Garcia-Pardina et al., 2024; Jimenez & Garrido, 2025).

The incorporation of network-based item filtering yielded notable improvements in EGA performance. Following redundancy detection with UVA and stability filtering with bootEGA, which together removed one-third to one-half of the generated items, EGA's performance approached near-ceiling levels across all evaluation criteria. Dimensionality estimation became highly accurate with bias effectively eliminated, correct structure identification occurred in the vast majority of replicates, and structural recovery reached near-perfect levels. These improvements illustrate how network psychometric methods address core challenges in generative AI psychometrics that extend beyond the scope of traditional dimensional structure estimation procedures.

The necessity of this filtering reflects fundamental differences between LLM-generated item pools and traditionally developed measures. When human experts craft scale items, they explicitly avoid redundant phrasing, prioritize conceptual distinctiveness within dimensions, and iteratively refine items through qualitative review processes before data collection (Gehlbach & Brinkworth, 2011). LLMs, by contrast, generate items through probabilistic sampling from learned language distributions, producing item pools that inherently contain paraphrastic variants, semantically overlapping expressions, and variable degrees of construct specificity (Fyffe & Kaplan, 2024; Lee et al., 2025; Russell-Lasalandra et al., 2024). Although this generative process is efficient and yields large item pools rapidly, it also introduces redundancy and potential structural instability that can confound dimensionality estimation when left unaddressed. The AI-GENIE network-based filtering pipeline systematically identifies and removes such problematic content through explicit detection of redundancy and instability, producing refined item pools whose embedding-based similarity structures more clearly represent the intended psychological constructs (Russell-Lasalandra et al., 2024).

Taken together, these findings provide clear evidence that current default practices in embedding-based psychometrics, specifically the application of PCA with traditional retention criteria to similarity matrices, yield systematically biased dimensional structure estimates and warrant reconsideration. The substantial overextraction observed even with parallel analysis suggests that the problem reflects fundamental incompatibilities between eigenvalue-based PCA procedures and the properties of similarity matrices derived from large LLM-generated item pools. By contrast, the EGA and AI-GENIE approaches demonstrated robust performance that improved further when integrated with specialized network methods for addressing item redundancy and structural instability.

Limitations

There are some limitations in this study that should be noted. First, the study relied on a single LLM (GPT-4o) for item generation and a single embedding model (OpenAI's text-embedding-3-small) to derive semantic representations. Different LLMs or embedding models may produce items or semantic spaces with differing characteristics, which could affect absolute levels of structural recovery. Nonetheless, the observed performance patterns closely aligned with well-established limitations of PCA-based methods documented across diverse psychometric contexts (Goretzko & Bühner, 2021; Samo & Christensen, 2025; Widaman, 1993). This convergence suggests that the relative superiority of network approaches is likely to persist across different combinations of generative and embedding models, even if absolute accuracy levels vary.

Second, the simulation held constant the number of dimensions, consisting of six facets nested within two domains, the target number of items per dimension (30), and the content area (personality). Accordingly, the findings are tied to these constraints, and caution is warranted when generalizing to structures with different characteristics or content domains. It should be noted, however, that these simulation specifications created challenging conditions representative of LLM-based workflows (Lee & Yao, 2023; Russell-Lasalandra et al., 2024), including relatively high dimensionality, large item pools, and hierarchical nesting.

Another limitation of the present study is that we did not compare network approaches against the full range of traditional dimensional structure estimation methods that have been applied to embedding-based representations. Although alternative methods to PCA, such as multidimensional scaling, k-means clustering, and factor or structural equation models have been used in embedding contexts, our comparison focused on PCA because it dominates current applications and because traditional factor analytic and clustering approaches also face well-documented challenges that network psychometric methods were explicitly developed to address (Samo & Christensen, 2025). Nonetheless, future research should systematically evaluate the performance of these alternative methods in embedding-based contexts to provide a more comprehensive methodological comparison.

Practical Implications

Our findings provide practical guidance for generative psychometric workflows in which item pools are produced by LLMs, represented in semantic vector spaces, and evaluated before human response data are available. In conventional exploratory practice, large item pools require proportionally large samples to stabilize covariance and dimensional structure estimates (Hirschfeld & Thielsch, 2014; MacCallum & Hong, 1999), creating a bottleneck for early prototyping. The present results indicate that embedding-based similarity matrices can be used as a pre-empirical screening step to examine the semantic organization of an item pool, remove redundant or unstable items, and refine construct specifications prior to data collection. This approach is consistent with recent proposals to integrate LLMs into test development while retaining expert oversight (Brickman & Oltmanns, 2025; Fyffe & Kaplan, 2024; Kowal & Kantrowitz, 2025; Lee & Yao, 2023; Terry & Waychoff, 2025). Importantly, however, the study shows that dimensional recovery in embedding space depends critically on analytic choices. Across simulations, PCA-based pipelines applied to similarity matrices severely overestimated dimensionality and yielded poorer structural recovery, whereas network psychometric workflows using EGA produced more accurate structural representations, particularly when combined with AI-GENIE filtering procedures based on UVA and bootEGA. Methodological choices at this pre-empirical stage are therefore foundational rather than preliminary, as dimensional structure estimation errors propagate through item selection, dimension definition, construct labeling, and downstream theoretical interpretation in ways that subsequent empirical data analyses may not be able to fully correct.

Beyond these procedural considerations, the study also reinforces that semantic structure should be treated as a substantive component of measurement rather than merely a convenient proxy for later psychometric structure. Emerging evidence suggests that item semantics can anticipate empirical item covariation and support cross-instrument prediction, implying that observed dimensional organization is partly constrained by wording and content decisions (Ravenda & Raballo, 2025). This perspective aligns with work arguing that LLM-based embeddings can quantify construct overlap, diagnose jingle–jangle problems, and address taxonomic incommensurability by locating items and construct labels within a shared semantic space (Wulff & Mata, 2025a, 2025b).

Embeddings permit taxonomic evaluation across multiple instruments and broad construct domains simultaneously, a scope of comparison that empirical validation cannot achieve because administering all candidate items together and collecting the necessary sample would be infeasible (Hirschfeld & Thielsch, 2014). In practical terms, embedding-based EGA, taxonomic graph analysis, and AI-GENIE pipelines (Russell-Lasalandra et al., 2024; Samo & Christensen, 2025) provide methodological tools for assessing content validity and conceptual clarity by identifying dimensions that are semantically indistinguishable despite distinct labels, detecting drift of item sets toward adjacent constructs, and revealing misalignment between construct labels and item content. By treating item semantics as an independent source of construct validity evidence rather than a proxy awaiting empirical confirmation, embedding-based workflows position measurement science to more directly address persistent taxonomic ambiguities that have remained difficult to resolve within purely response-based validation frameworks.

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